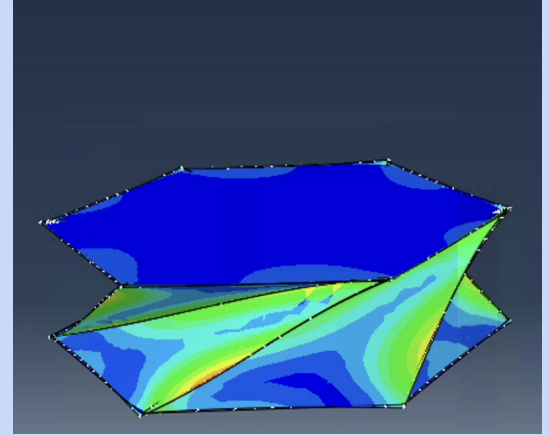
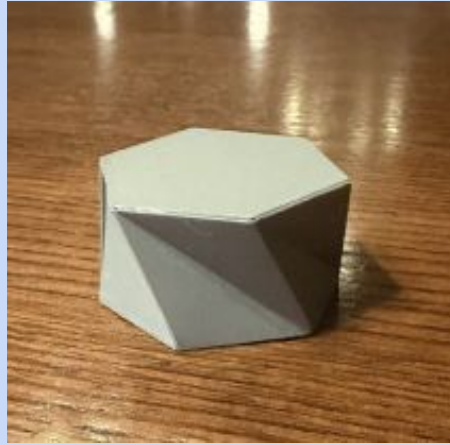
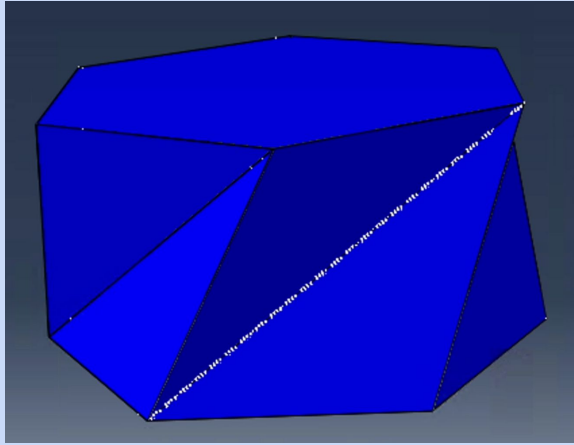




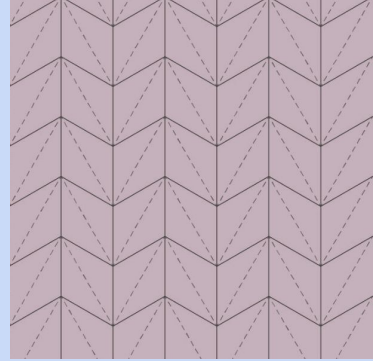
Origami Kresling Unit Cell for Abaqus CAE

Ella McRitchie, Anna Burgess

Origami Kresling Background

- Cylindrical origami shell based on the Kresling pattern
- Intrinsic coupling between axial displacement and rotational motion
- Modular unit cell that can be easily combined to form multi-cell structures
- Relevant applications include soft robotics, deployable systems, energy-absorbing components, and morphing mechanisms

Why a FEA Model is valuable: Optimize geometry for stiffness, energy absorption, and actuation with different parameters that would otherwise be extremely time consuming to test and fabricate correctly



Our Goal

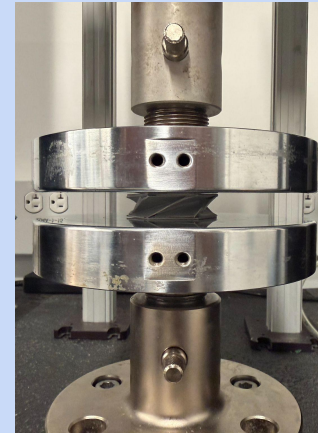
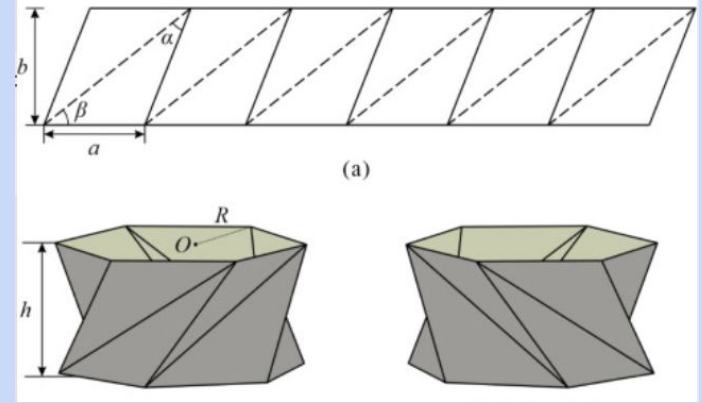
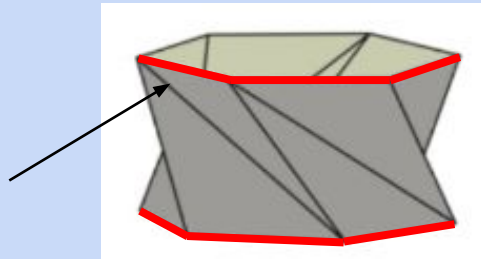
Model a kresling in abaqus that is synonymous with both the Behavior of our physical model and the physical properties of our model

Our
Approach



Create Physical Model

- 4 cm side lengths
- 4 cm height
- ABS filament
 - Young's Modulus: 2.08×10^9
 - Poisson Ratio: 0.35
- Double sided tape connecting triangle to hexagon faces to try and reduce compliance issues
- 3D printed at a material thickness of 0.4 mm, and a material thickness of XX mm at the creases.



Compression Test

Create Abaqus Model

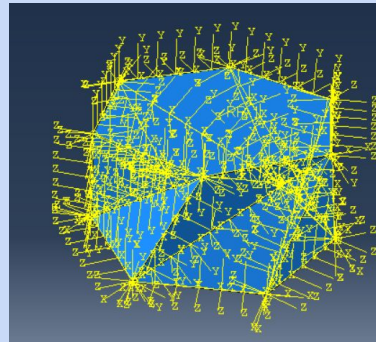
- Hexagon top/bottom are unique parts with steel properties for the material settings, representing the compression plates of the Intron
- Triangular Facet parts are Assigned ABS properties
- Mesh Size: 0.07 m
- Element Type: S3
- Mesh Controls: Tri, Free
- Hexagon and triangle parts brought together with an instance on the assembly, but not merged, such that separate nodes are retained between parts with shared edges.
- BCs: Encaster the bottom edge, Displace a reference point that is coupled to the top edge

```
# Material (ABS)
E_ABS      = 2.08e9  #Pa
nu_ABS     = 0.35

# Material (Steel)
E_steel    = 200e9  #Pa
nu_steel   = 0.3
```

Creating Hinge Connections

- We found matching node pairs between two edge sets and determined the hinge direction.
- At each node we built a local coordinate system for consistent hinge orientation.
- Created six springs per node pair: 3 linear, 1 rotational hinge, and 2 rotational locks. The pairs were defined by every edge in the assembly.
- Iterated through all hinge definitions to generate the full spring-based hinge network.



Datum Axis for springs

```
# 1) Bottom circumferential hinges: hexbottom#a <-> bottomtri#b
for i in range(N):
    set1 = 'hexbottom%d' % (i+1)
    set2 = 'bottomtri%d' % (i+1)
    hname = 'hb%02d' % hid
    hinges.append((set1, set2, hname, True))
    hid += 1

# 2) Top horizontal hinges: tophex# <-> toptri#
for i in range(N):
    set1 = 'tophex%d' % (i+1)
    set2 = 'toptri%d' % (i+1)
    hname = 'ht%02d' % hid
    hinges.append((set1, set2, hname, True))
    hid += 1

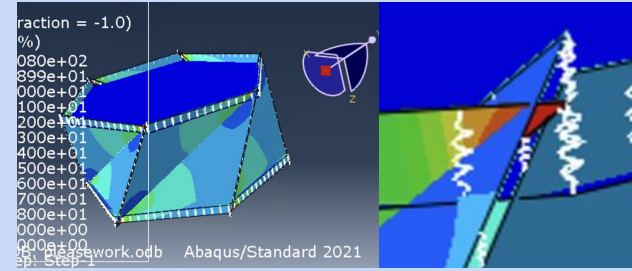
# 3) Vertical panel-panel edges: vertedge#a <-> vertedge#b
for i in range(N):
    set1 = 'vertedge%d'a' % (i+1)
    set2 = 'vertedge%d'b' % (i+1)
    hname = 'vv%02d' % hid
    hinges.append((set1, set2, hname, False))
    hid += 1

# 4) Diagonal panel-panel edges: triedge#a <-> triedge#b
for i in range(N):
    set1 = 'triedge%d'a' % (i+1)
    set2 = 'triedge%d'b' % (i+1)
    hname = 'td%02d' % hid
    hinges.append((set1, set2, hname, False))
    hid += 1

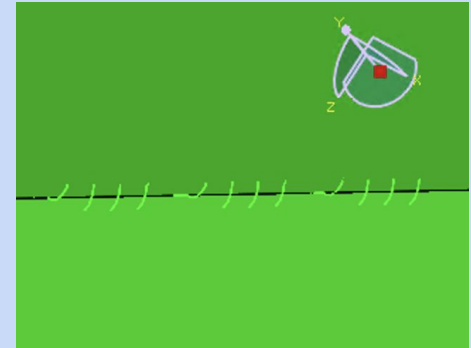
# 5) Circumferential extra hinges: circedge#a <-> circedge#b
for i in range(N):
    set1 = 'circedge%d'a' % (i+1)
    set2 = 'circedge%d'b' % (i+1)
    hname = 'cc%02d' % hid
    hinges.append((set1, set2, hname, False))
    hid += 1
```

Simulation Path to Success- Sticking Points

- Started with one spring for every edge and a merged model- this didn't run at all
- Now one spring for every node along the edge with unmerged panels
 - Spring/Dashpot Elements
 - Springs exist between every boundary and are made by tying together sets, where each edge in the model has two sets- one for each instance that has a shared edge in the unmerged model
- Step: Static General
 - NLGEOM on
 - Automatic Stabilization: Energy dissipation at $2e-7$



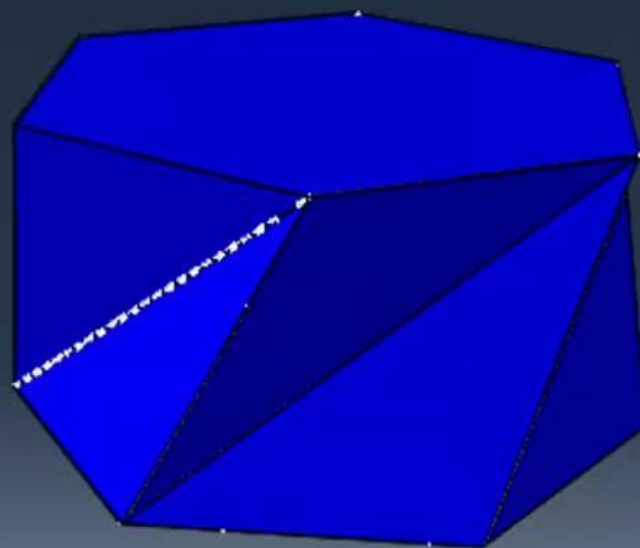
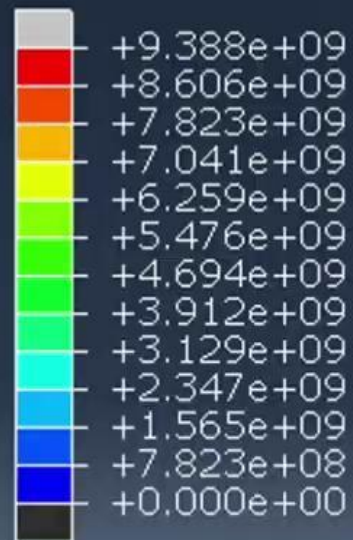
Compliance issues as our unmerged model gave no room between panels



Correctly Orientated Springs

Our Final Working Model

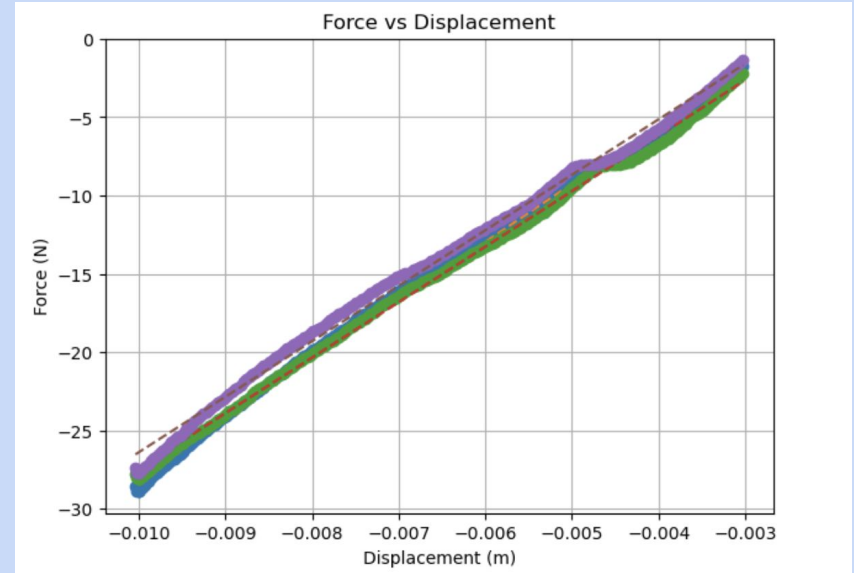
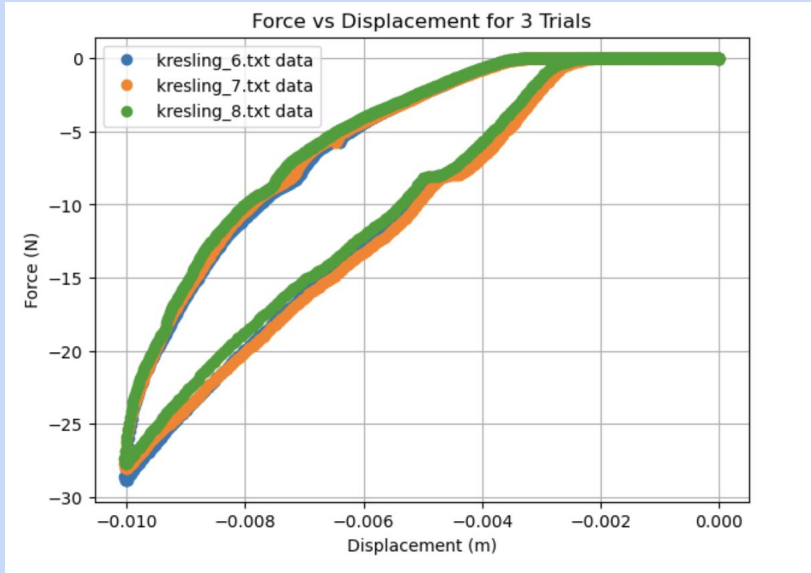
S, Mises
SNEG, (fraction = -1.0)
(Avg: 75%)



Step: Step-1 Frame: 6
Total Time: 0.014187



Comparison Between Physical Model and Simulation

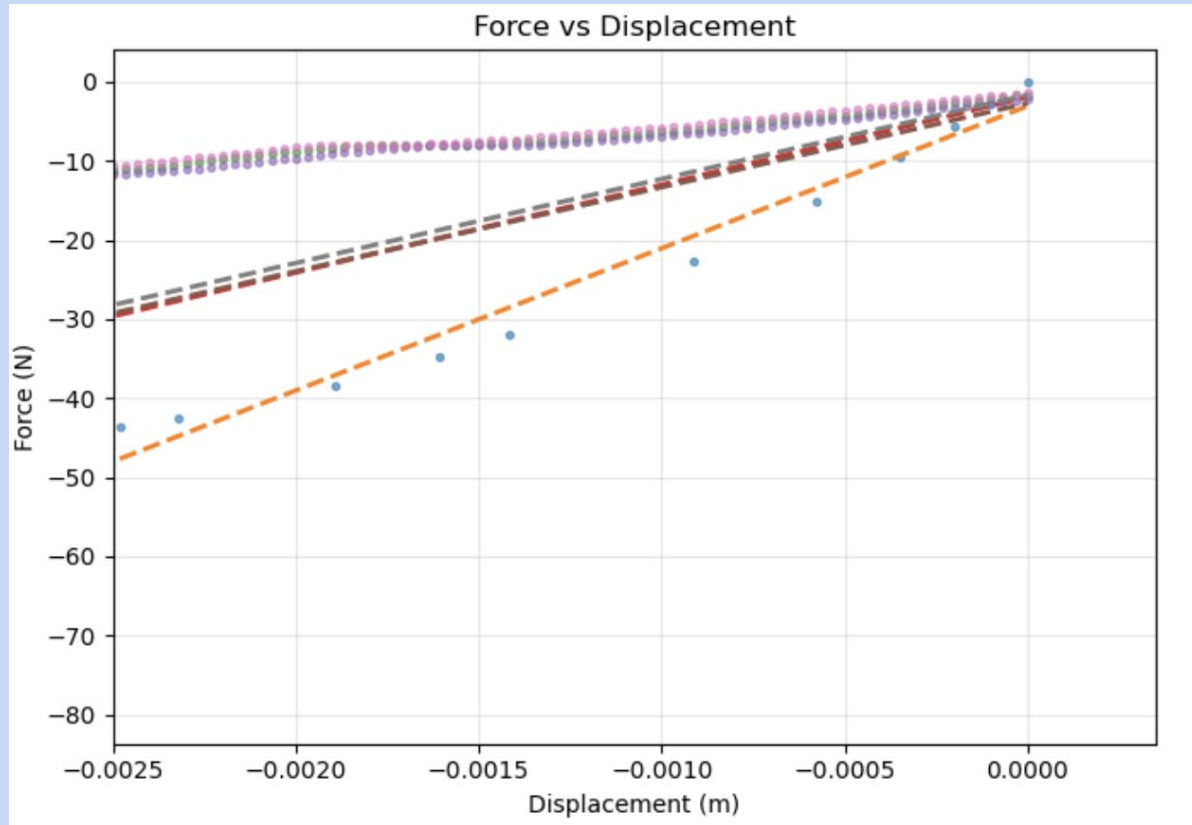


Spring Constant (Divide Found Value by 12 springs in parallel):

- Physical Model 885 N/mm
- Simulation Model 906 N/mm

```
# Spring Parameters
k_rot = 0.0009
k_rot_lock = 1000.0
k_lin = 1.0e10
TOL = 0.05
```

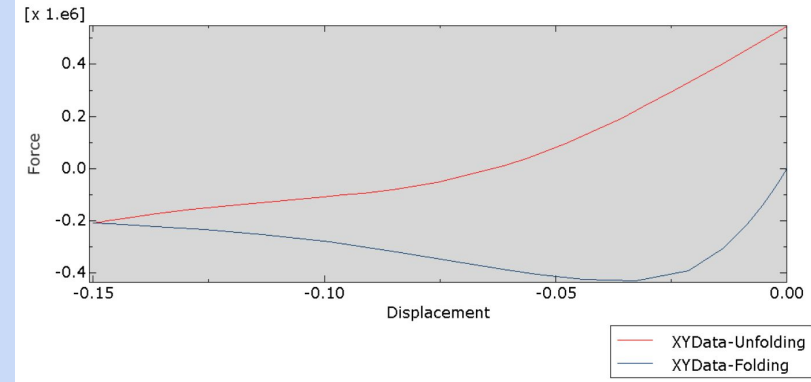
Behavior Comparison



Hysteresis and Plasticity in Our Model

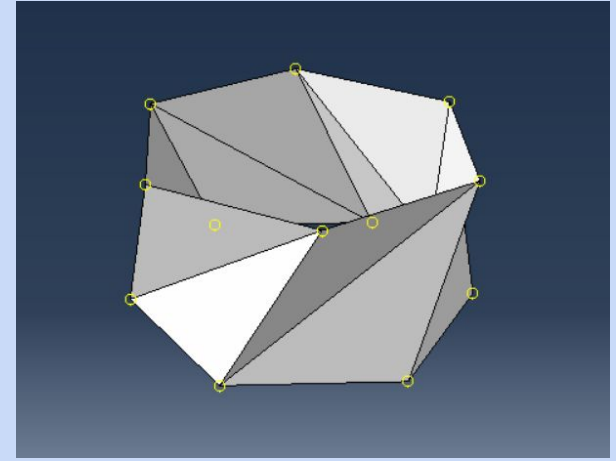
- Physical tests showed hysteresis, but the initial Abaqus model did not.
- Adding an unfolding step produced identical loading/unloading curves → no hysteresis.
- Adding plasticity introduced hysteresis, but caused excessive permanent deformation after one cycle.
- Future improvements for capturing hysteresis more accurately:
 - Using nonlinear contact or frictional hinge behavior
 - Geometric imperfections

```
# Plasticity Parameters
yield_point = 35e6
hardening = 50e6
near_ultimate = 55e6
softening = 45e6
yield_point_percent = 0.00
hardening_percent = 0.0260
near_ultimate_percent = 0.0736
softening_percent = 0.1784
```



Full Parameterization of Model

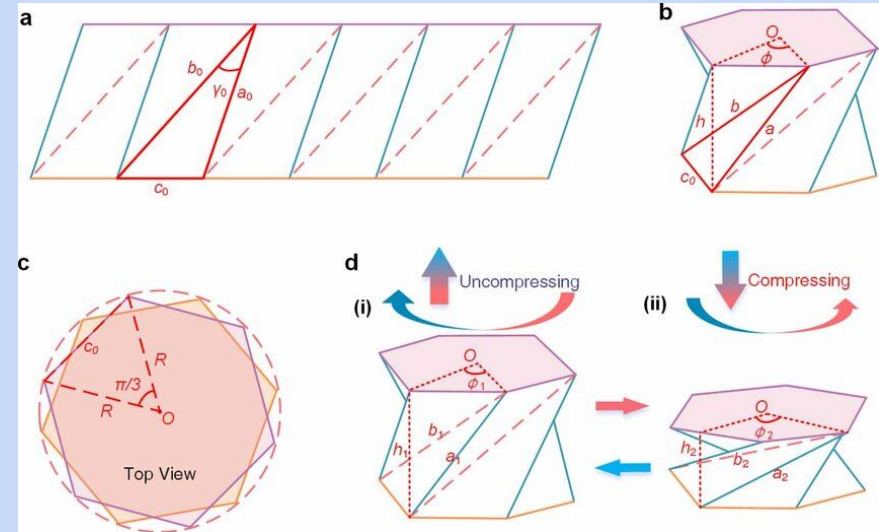
- Reconstructed the model geometry to improve parameterization.
- Each part is generated by defining datum points at coordinates determined by:
 - R – radius of the polygon
 - N – number of polygon sides
 - H – height of the unit cell
- Although the full unit cell could be created in a single part, we intentionally kept multiple parts assembled as an instance- ensures separate nodes along edges, enabling insertion of spring elements where required.



Datum Point Driven Model

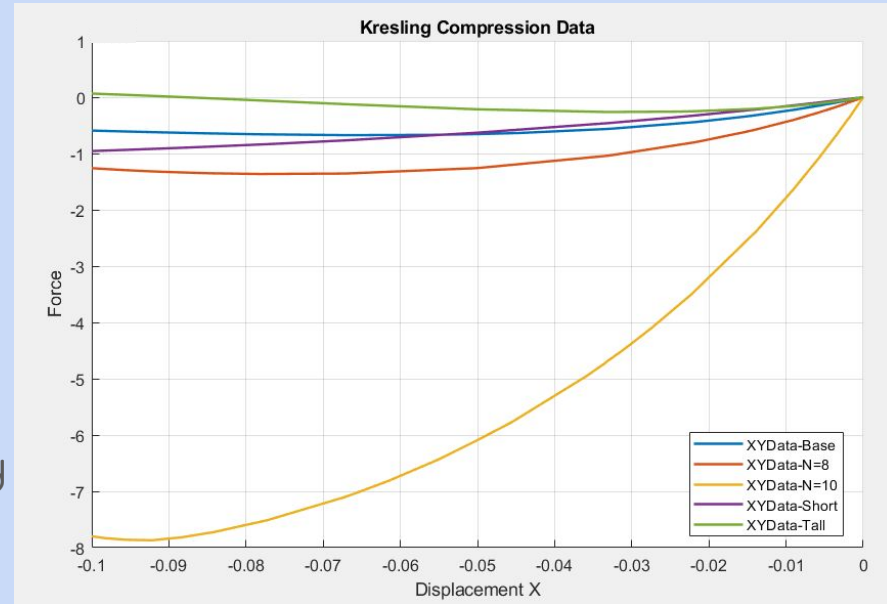
Geometry Parameterization

- $\theta = 2\pi i/6$ ($i = 1, 2, 3, 4, 5, 6$)
- $\phi = 30(\pi/180)$
- Bottom Hexagon Datum Points:
 - $x = R\cos(\theta)$, $y = R\sin(\theta)$
- Top Hexagon Datum Points:
 - $x = R\cos(\theta + \phi)$, $y = R\sin(\theta + \phi)$
- Wire features are created by connecting these datum points.
- A shell surface is formed using the cover edges tool.

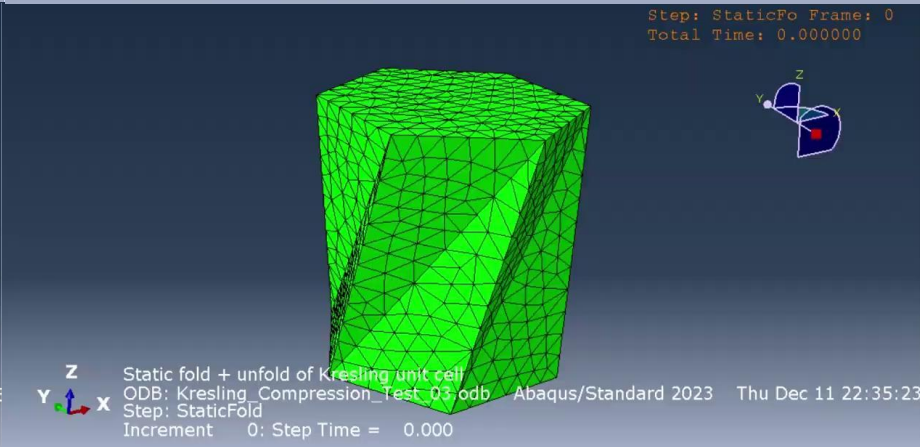
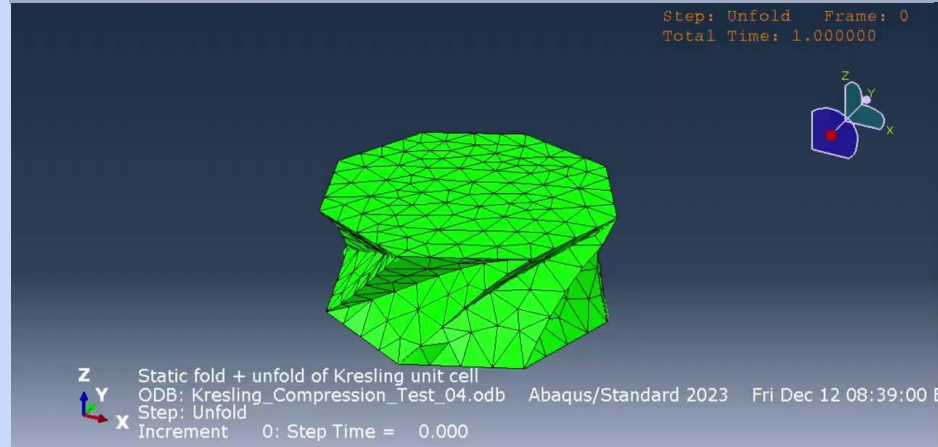
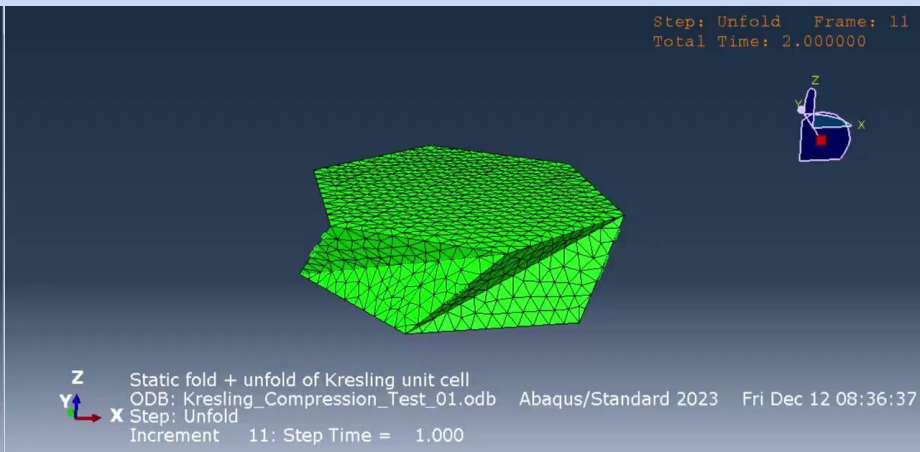
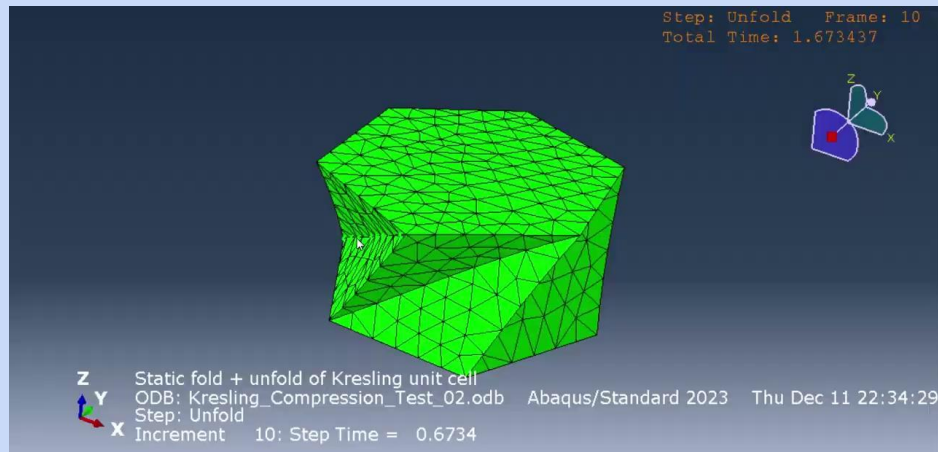


Force Curves for Varying Model Geometry

- More sides ($N=8$, $N=10$) → smoother deformation and slightly reduced stiffness, producing lower reaction forces for the same displacement.
- Shorter height → structure becomes stiffer, showing higher compressive force because panels rotate less and load is carried more directly.
- Taller height → structure becomes much softer, allowing large rotation and inward folding, which sharply reduces initial stiffness and delays force buildup.



Different Model Simulations



New Skills and General Takeaways

- Debugging and troubleshooting strategies
- Think about how to work with flaws of modeling instead of always completely avoiding them (energy dissipation)
- In general, finding simpler set-ups to perfect mechanisms before applying them to much more complex geometries like the kresling speeds up the design process (i.e. hinging two panels together before moving starting to hinges on the unit cell)